

IBM HACKATHON PROJECT

Aarogyam: AI-Powered Health Literacy App.

Presented By:

Faculty name : Neha G. Rathi

Student Name_1 : Pranav Pradip Raut

Student Name_2 : Prit Nilesh Thombare

College Name : Sipna College Of Engineering and Technology, Amravati (CSE)

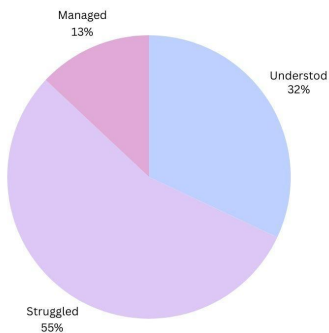
Department : Computer Science and Engineering

OUTLINE

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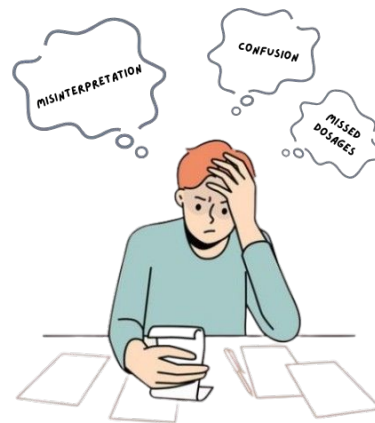
PROBLEM STATEMENT

A key challenge in health care management is **understanding complex medical reports and prescriptions**, which can lead to misinterpretation, incorrect medication administration, and a **lack of health literacy**. These issues can result in **missed doses** and **inaccurate diagnoses**, ultimately compromising the **quality of care**.



- **32%** of surveyed individuals fully understood the information provided in their patient information leaflets.
- **55%** struggled with reading or comprehending the details.
- **13%** faced some difficulty but could partially manage.

Source



Aarogyam tackles this by:

- Simplifying medical information.
- Providing automated reminders.
- Treatment options like Ayurveda and yoga.
- Local health insights.
- multilingual support.

TECHNOLOGY USED

Platform:

- **Android Studio:**
The development environment used to build and test the app.
- **Render:**
Hosting platform for deploying backend services.

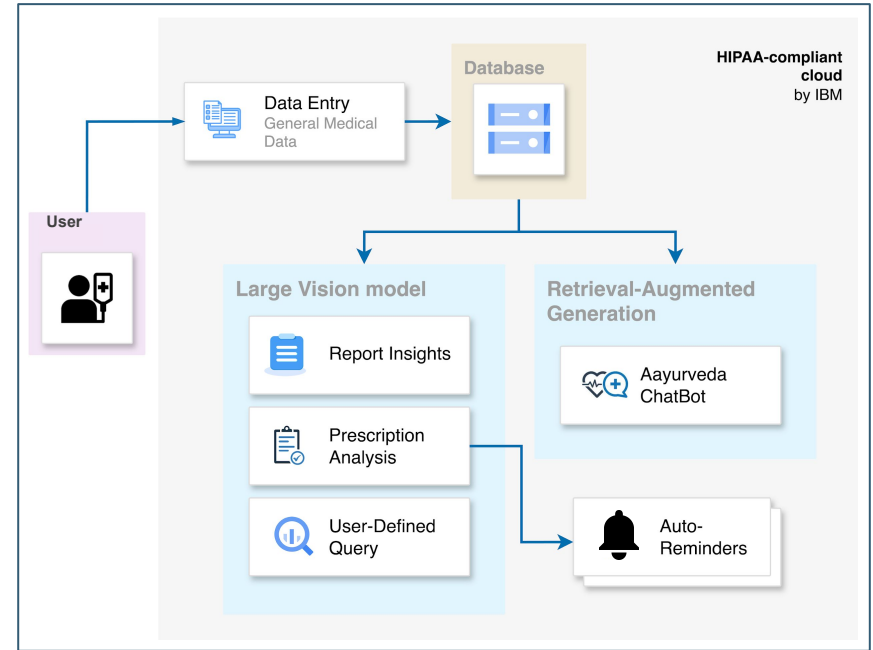


Technologies:

- **Flutter:**
Used for creating the app's responsive and cross-platform user interface.
- **Dart:**
The programming language for building the app's logic and frontend functionality.
- **IBM Cloud:**
Utilized for storing data, integrating AI features, and processing user inputs securely.
- **SQLite:**
Local database for storing user preferences, reminders, and offline data.
- **Firebase:**
Used for user authentication and real-time syncing of reminders and settings.



Architecture



IBM CLOUD SERVICES USED



WOW FACTORS

Simplified Health Literacy:

Turns complicated medical reports and prescriptions into layman's terms.

Ayurvedic Chatbot:

Unique traditional remedies and remedial guidance mapped to the modern world.

Smart reminders:

Auto reminders are set once the prescription is processed.

Multilingual:

Supports Marathi, Hindi, and English so that we can address the myriad of user needs.

User-specific outputs:

The output is formatted for the users who gave allergies, age, gender, etc.

Easy to use:

Designed with a user-friendly and versatile interface for users across the age.

END USERS

Non-medical Professional:

People without medical literacy need easy access to information.

Caregivers:

Family managing the health of their loved ones.

Patients

People who need to be re-explaining their medical test results or prescriptions in simpler words.

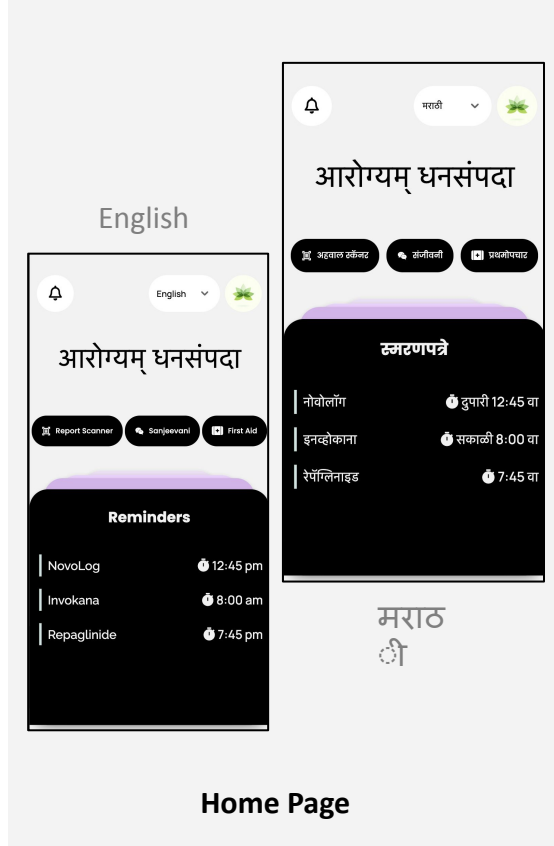


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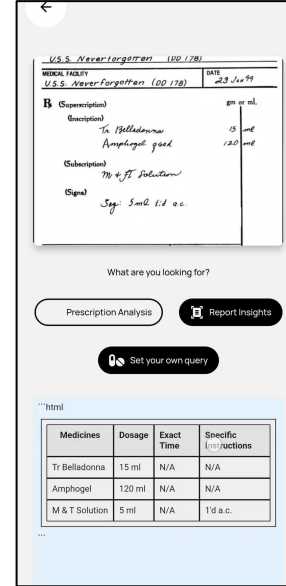
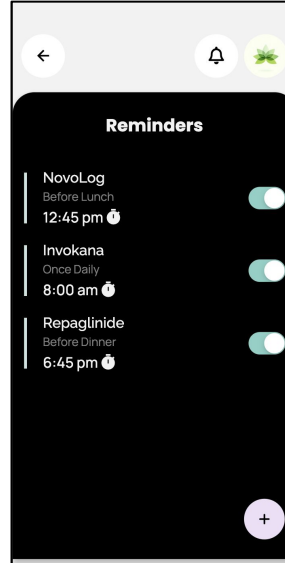
Health Enthusiast on Self-Care:

Persons looking for ayurvedic solutions or health tips according to them.

RESULTS

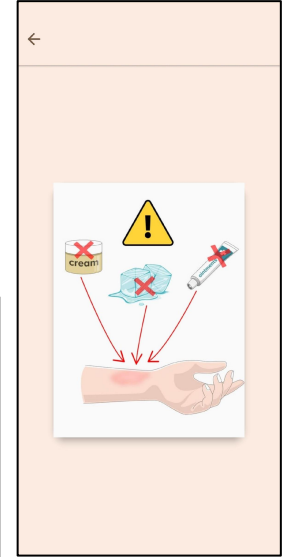
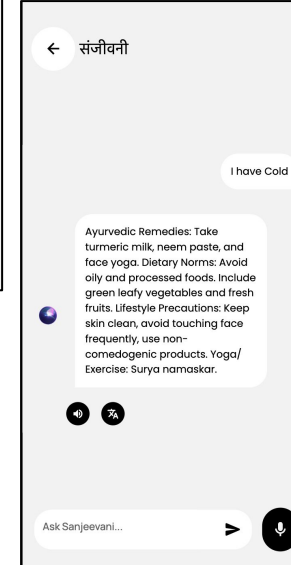


Automatic Reminders



Prescription Analysis

Ayurveda ChatBot



First-Aid

CONCLUSION

Linking Complexity to Clarity

Aarogyam fills the significant health literacy gap by simple language understanding concepts and the treatment plan designed for the patients.

It allows people to take care of themselves and gives control and confidence in healthcare.



Simplifying Healthcare for All

Being medicine presents Ayurveda remedies along with customised advice and reminders that make heal multilingual (Marathi, Hindi and English), and client-centricity focuses on the problems stated within the problem statement of Aarogyam — simplifying complex medical info for patients.

The app also simplifies the care approachable and easy for every individuals

Addressing the Constraints in Problem Statement:

Aarogyam: While dealing with the issue itself, Aarogyam tackles this as re-framing and re-personalizing the interaction a person has with their health.

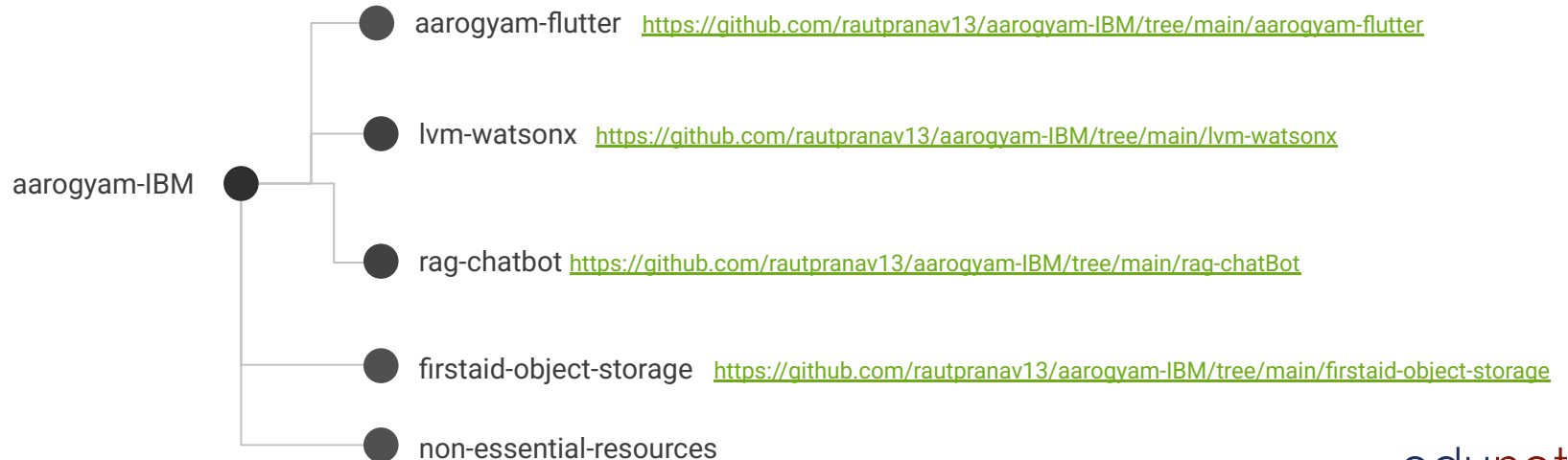
It turns healthcare from a large blob of chaos and confusion to focused, understandable for the user.



GITHUB LINK

■ <https://github.com/rautpranav13/aarogyam-IBM>

Project Structure:



FUTURE SCOPE

- **Advanced Symptom Checker**

Use AI to study symptoms and figure out what they might be pointing to.

- **Personalized Disease Risk Assessment:**

Provide risk scores specific to your user given available data.

- **Wearables Integration:**

Integrate health devices to have real-time data and insights.

- **Disease Prediction:**

Take Audio-visual input to predict certain diseases.

- **Tele-communication:**

Enable direct communication with health professionals for unresolved queries.

- **Language Expansion:**

Provide risk scores specific to your user given available data.

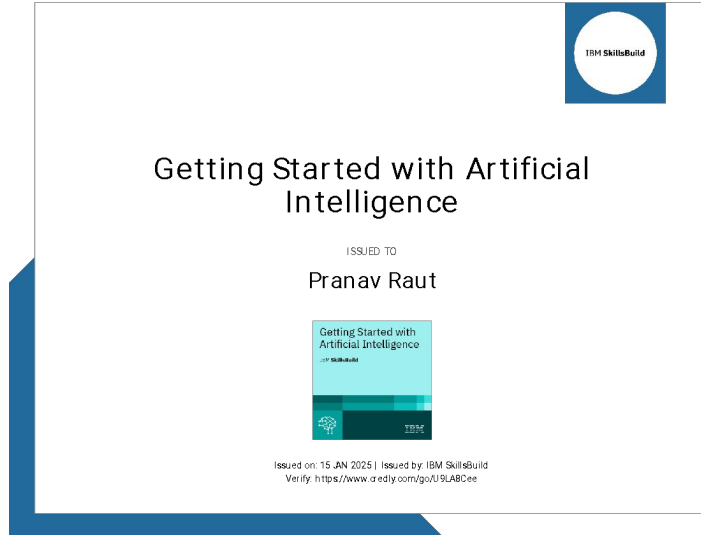
- **Real-time Allergy Alerts:**

Integrate health devices to have real-time data and insights.

- **Auto-generated preventive health plans:**

Take Audio-visual input to predict certain diseases.

IBM CERTIFICATIONS





THANK YOU